

Advanced Web Design and Development (BIT3208) - Project Logbook

Developer: Kariuki Joseck

Project: Full-Stack User Authentication & CRUD System

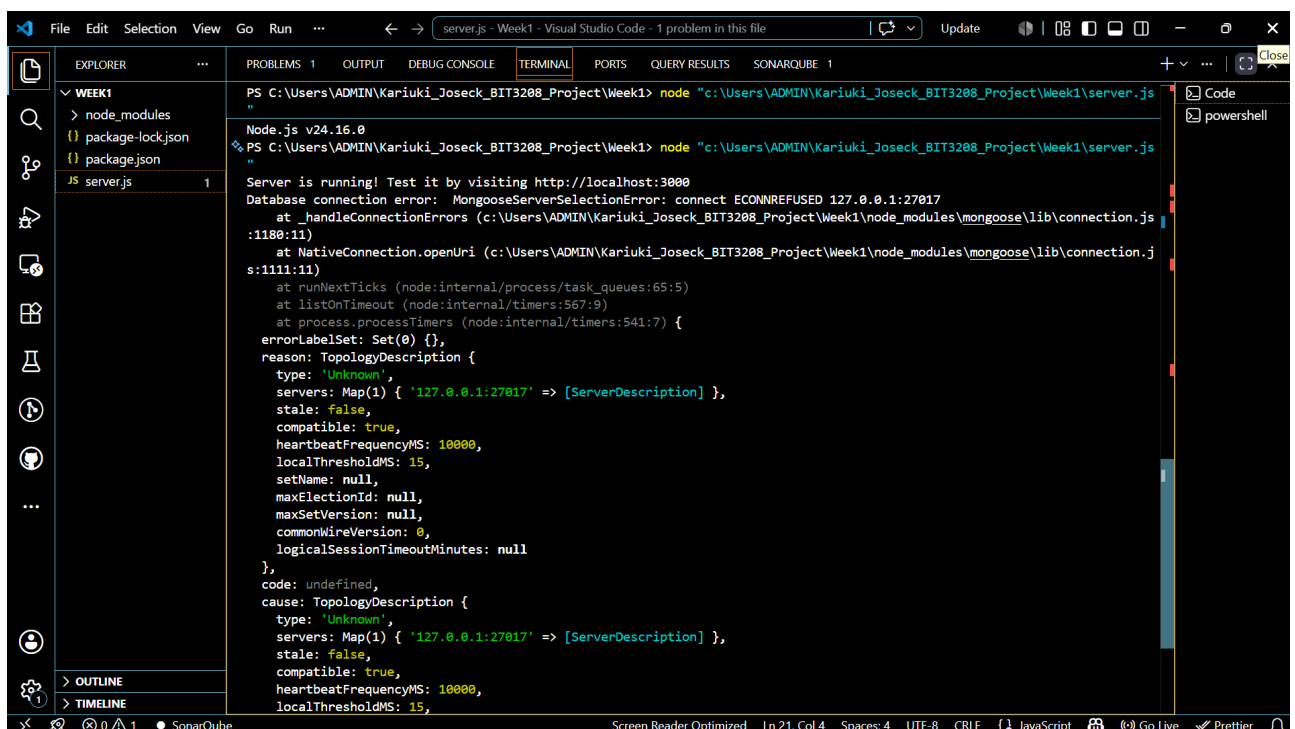
System Overview

A Node.js and Express backend application utilizing MongoDB for persistent data storage. The system features secure user registration, session-based authentication, and full CRUD (Create, Read, Update, Delete) capabilities.

Weekly Development Log

Week 1: Environment Setup

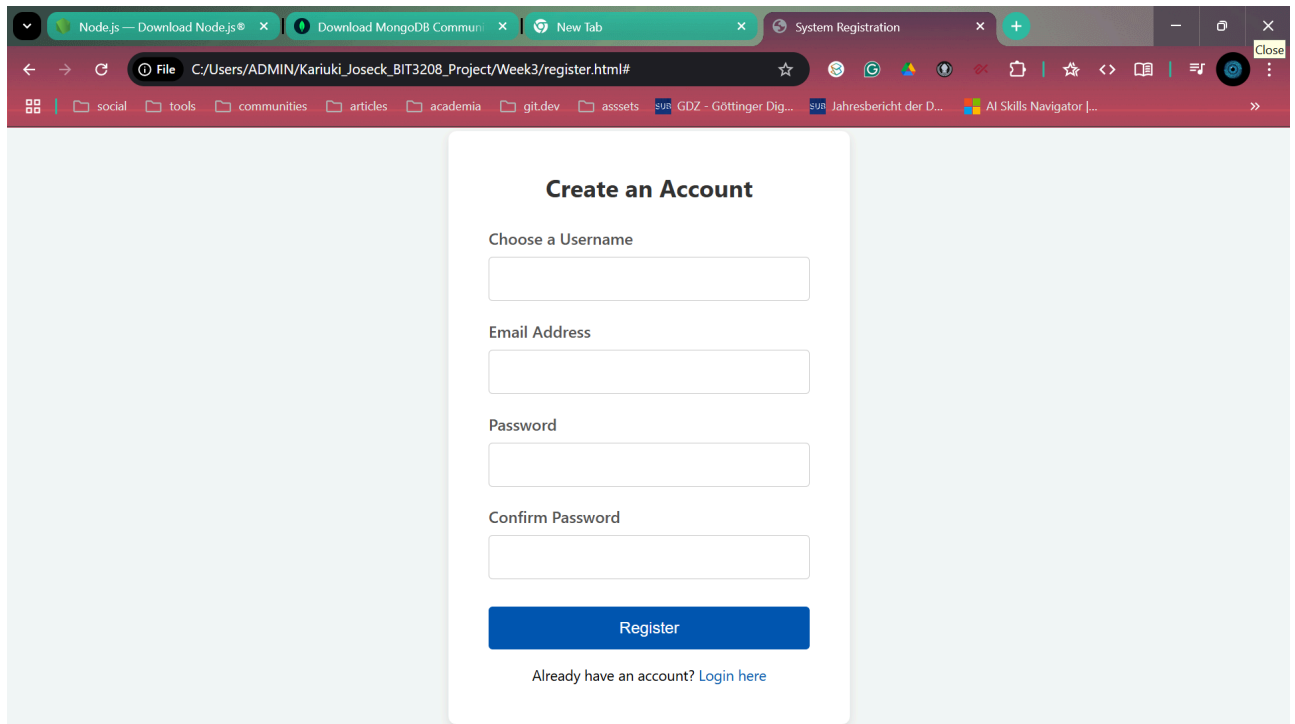
- **Objective:** Establish the local development environment and test core server capabilities.
- **Tasks Completed:** * Installed Node.js and MongoDB.
 - Successfully initialized the project environment and tested localhost connectivity.
- **Evidence:**

A screenshot of the Visual Studio Code interface. The Explorer pane on the left shows a file structure for 'WEEK1' with files 'node_modules', 'package-lock.json', 'package.json', and 'server.js'. The Terminal pane on the right shows the command 'node "c:\Users\ADMIN\Kariuki_Joseck_BIT3208_Project\Week1\server.js"' being executed. The output shows the server starting but then crashing with a 'MongoServerError: connect ECONNREFUSED 127.0.0.1:27017' error. The error details include a 'TopologyDescription' object with 'servers: Map(1) { '127.0.0.1:27017' => [ServerDescription] }', 'stale: false', 'compatible: true', 'heartbeatFrequencyMS: 10000', 'localThresholdMS: 15', 'setName: null', 'maxElectionId: null', 'maxSetVersion: null', 'commonWireVersion: 0', and 'logicalSessionTimeoutMinutes: null'. The status bar at the bottom indicates 'Screen Reader Optimized', 'Ln 21, Col 4', 'Spaces: 4', 'UTF-8', 'CRLF', 'JavaScript', 'Go Live', and 'Prettier'.

Week 2: Frontend Design & Planning

- **Objective:** Design the graphical user interface and plan the system architecture.
- **Tasks Completed:**
 - Designed interface wireframes.
 - Planned the request-response workflow between the client and the Express server.

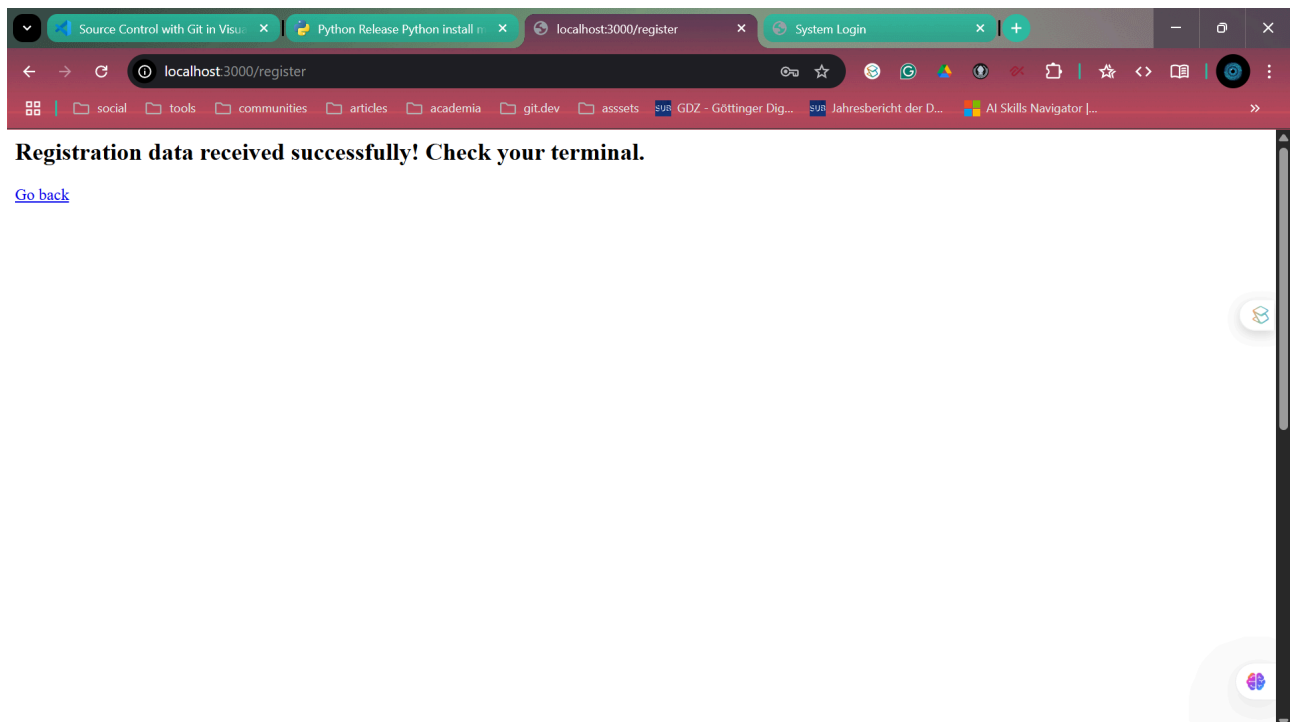
- **Evidence:**



The screenshot shows a web browser window with the address bar displaying 'C:/Users/ADMIN/Kariuki_Joseck_BIT3208_Project/Week3/register.html#'. The page content is a registration form titled 'Create an Account'. The form includes four input fields: 'Choose a Username', 'Email Address', 'Password', and 'Confirm Password'. Below these fields is a blue 'Register' button. At the bottom of the form, there is a link that says 'Already have an account? Login here'.

Week 3: Client-Side Logic & Form Validation

- **Objective:** Implement client-side interactivity and secure input handling.
- **Tasks Completed:**
 - Wrote JavaScript for DOM manipulation.
 - Implemented front-end form validation to ensure data integrity before server submission.
- **Evidence:**

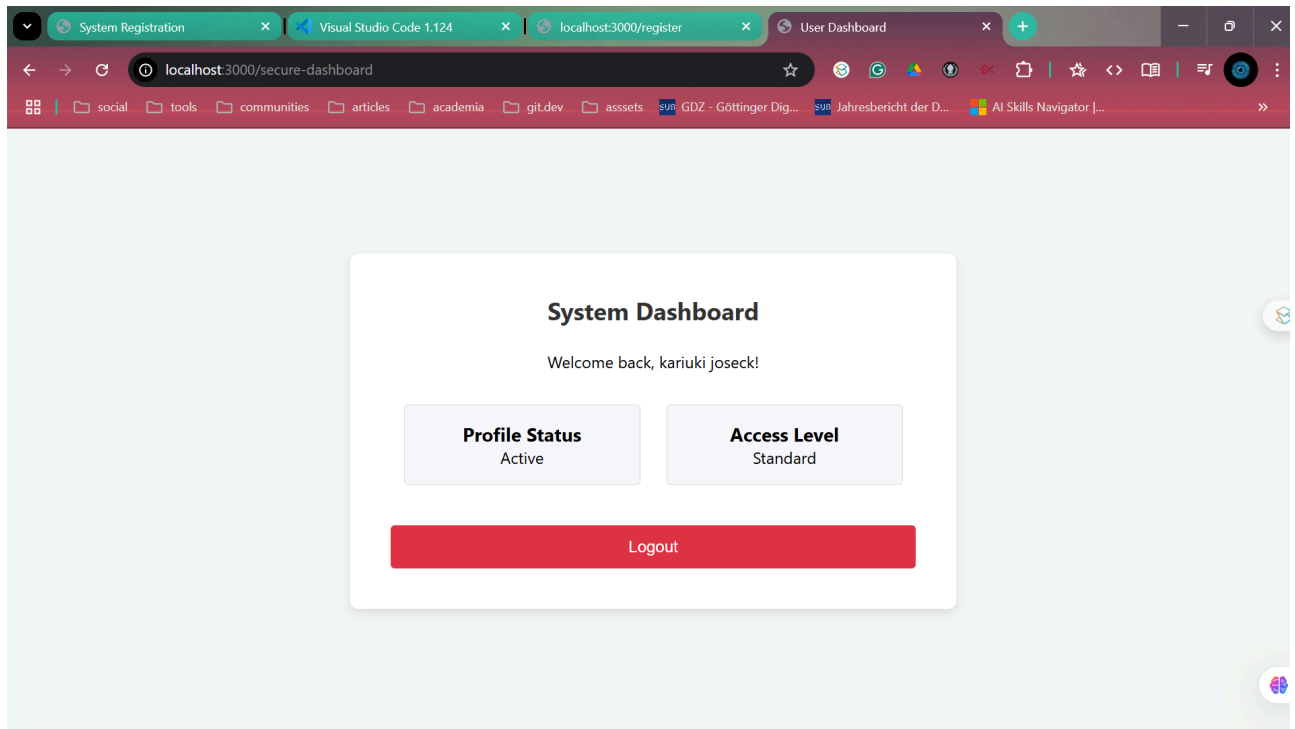


Week 4: Server-Side Processing & Authentication

- **Objective:** Build the backend processing logic and secure authentication pipeline.
- **Tasks Completed:**

- Configured Express server routing.
- Processed POST requests for user registration and login forms.
- Implemented secure session-based authentication to protect dashboard access.

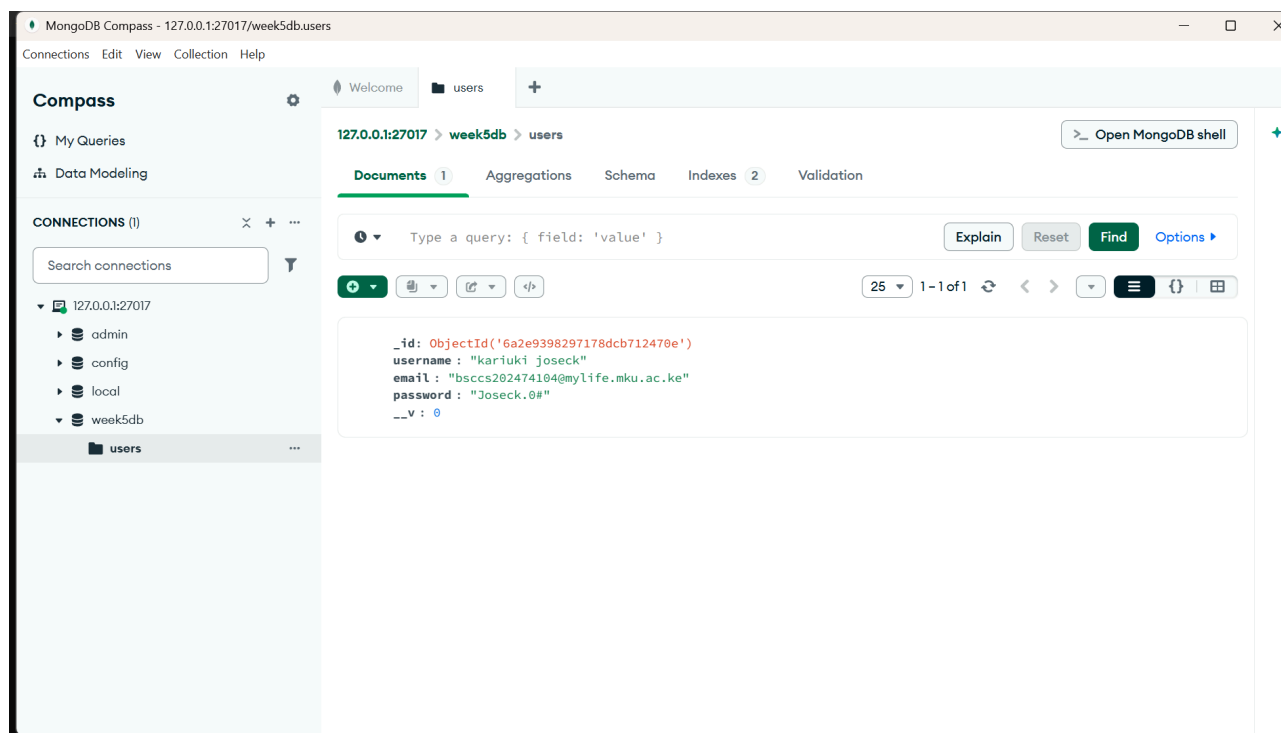
- **Evidence:**



Week 5: Database Integration & CRUD Operations

- **Objective:** Integrate a NoSQL database for persistent data storage and manipulation.
- **Tasks Completed:**
 - Connected Express server to MongoDB using Mongoose.
 - Created the foundational User schema.
 - Implemented full CRUD functionality: Create (registration), Read (login verification), Update (password modification), and Delete (account removal).
- **Challenges & Solutions:** * Addressed initial database environment setup error by identifying and reinstalling the correct version of MongoDB.

- **Evidence:**



Conclusion

The project successfully demonstrates a complete full-stack workflow, transitioning from static web interfaces to a dynamic, data-driven system using modern backend engineering practices and continuous version control via Git.

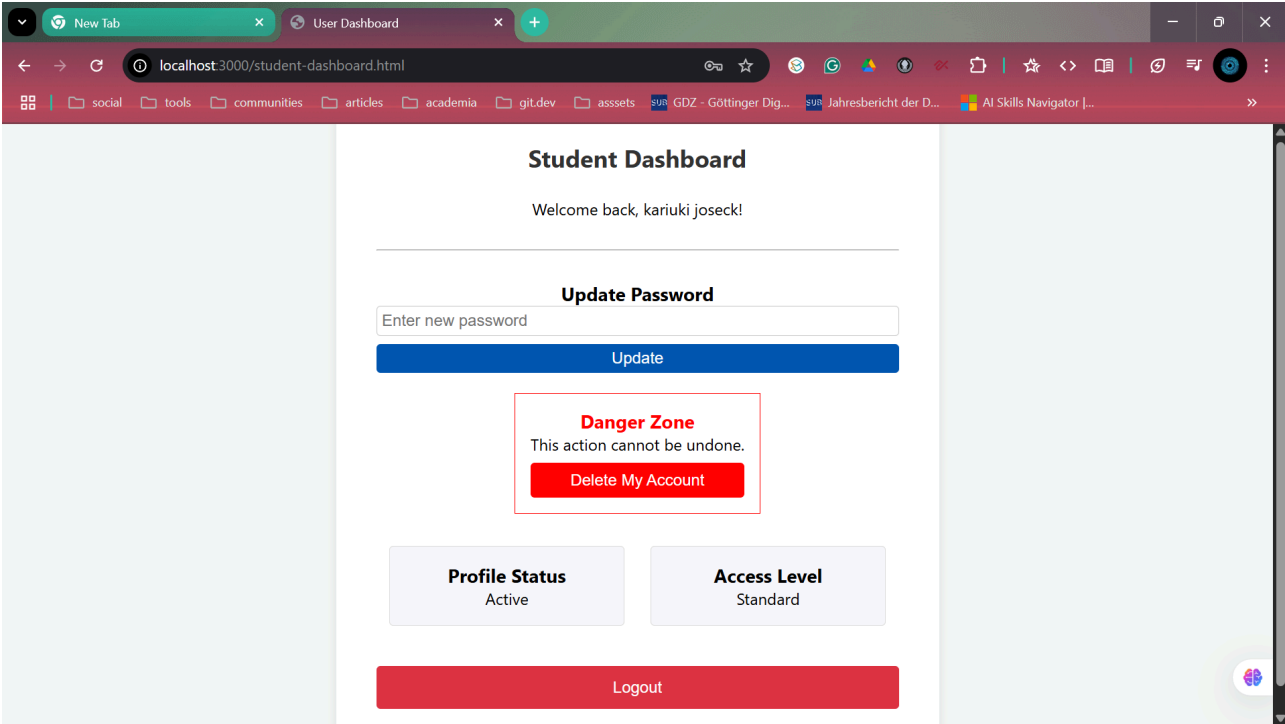
Week 6: Database Integration and CRUD Operations

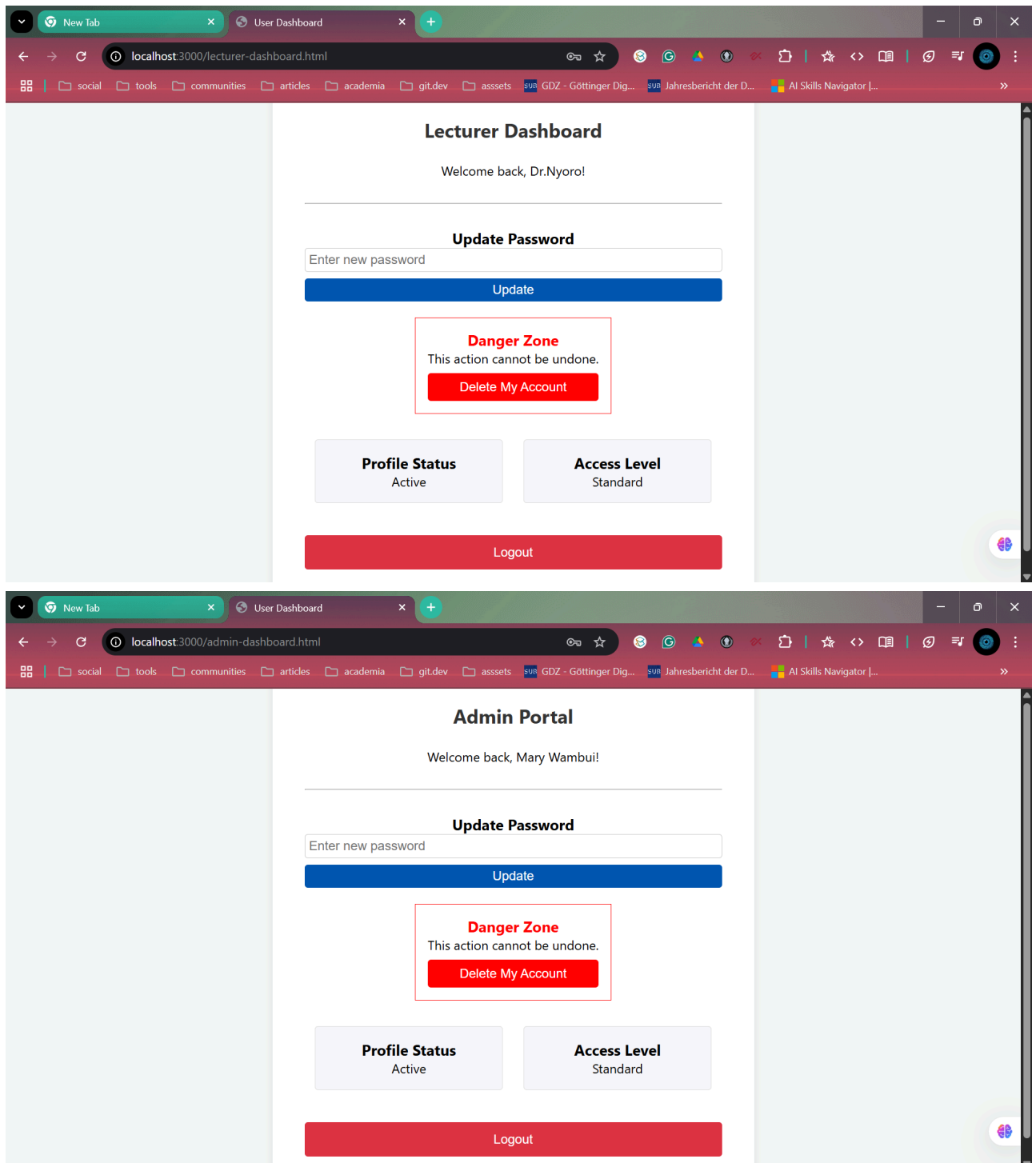
- **Objective:** Connect the web application to a database to persistently store and manipulate user data.
- **Tasks Completed:**
 - Configured a local MongoDB connection using the Mongoose ODM ([week6db](#)).
 - Designed the primary [User](#) model to structure incoming registration credentials.
 - Implemented backend routing for all CRUD operations: Create (registration), Read (login verification), Update (password modification), and Delete (account removal).
 - Authored a [database-schema.txt](#) document outlining the NoSQL architecture and data workflow.

Week 7: User Authentication & Role-Based Access Control

- **Objective:** Secure the application with encrypted passwords, session management, and multi-user routing.
- **Tasks Completed:**
 - Installed [bcryptjs](#) and implemented password hashing for all new registrations.
 - Upgraded the MongoDB User schema to include [Student](#), [Lecturer](#), and [Administrator](#) roles.
 - Built protected routes that verify active sessions before serving dashboard pages.
 - Implemented secure logout functionality to destroy sessions.

Evidence:





Week 8: Responsive Web Design and Mobile-First Development

- **Objective:** Apply Mobile-First design principles, CSS Flexbox, and CSS Grid to create highly adaptable, multi-device frontend interfaces.
- **Tasks Completed:**
 - Designed a responsive Personal Profile interface utilizing fluid CSS Flexbox containers.
 - Developed a dynamic Product Showcase using CSS Grid with specific tablet and desktop media queries.
 - Architected a complete 5-page responsive website ("Sweet Creations Bakery") utilizing a reusable global layout shell and responsive navigation.
 - Implemented mobile-first CSS scaling to ensure zero horizontal overflow and seamless rendering across smartphones, tablets, and wide-screen monitors.

- **Evidence:**

